

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – DEBUGGING & OPTIMISATION TASK

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Debugging & Optimisation Task
Weightage:	10% of CIA	Total Marks:	100

CO2 Statement: Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)

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Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given program/code segment before identifying errors.
- Clearly identify logical, syntactical, and performance-related issues in the code.
- Apply systematic debugging techniques to correct the errors effectively.
- Optimize the given solution by improving efficiency, reducing unnecessary computations, or enhancing time complexity wherever possible.
- Provide proper justification for all corrections and optimization decisions.
- Write clean, structured, and well-documented code with appropriate comments.
- Maintain clarity, logical reasoning, and technical accuracy throughout the solution.

Question Type	Focus Area	Questions
Debugging & Optimisation Task	Identify errors, debug programs, and optimize algorithm efficiency using appropriate techniques	Q1

SECTION A – DEBUGGING & OPTIMISATION TASK

Code of Conduct

I **V RAJESH KUMAR / 192511389** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Solutions for Q69 and Q88

Q69 – Debugging Maximum–Minimum Finder

Issue:

The loop starts from index 0 even though mn and mx are already initialized with arr[0]. This causes an unnecessary comparison and fails to handle an empty list.

Correct Code:

```
def min_max(arr):
    if not arr:
        return None
    mn = arr[0]
    mx = arr[0]
    for i in range(1, len(arr)):
        if arr[i] < mn:
            mn = arr[i]
        if arr[i] > mx:
            mx = arr[i]
    return mn, mx
```

Time Complexity: O(n)

Space Complexity: O(1)

Q88 – Debugging Pair Sum Code

Issue:

The outer loop stops too early using range(len(arr)-2), so the last valid outer index is skipped.

Correct Code:

```
def pair_sum(arr, target):
    for i in range(len(arr) - 1):
        for j in range(i + 1, len(arr)):
            if arr[i] + arr[j] == target:
                return (i, j)
    return None
```

Time Complexity: O(n²)

Space Complexity: O(1)